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BIODIVERSITY: Introduction

Biodiversity, or biological diversity, refers to the vast array of species of plants, animals, and microorganisms that live on Earth, as well as the ecosystems and genetic diversity within those species. It also includes the interactions between different species and their environments.

Biodiversity is a key component of a healthy and thriving planet, and it provides many benefits, including:

- Ecosystem services like clean air and water, soil formation, and climate regulation
- Food security and nutrition
- Medicine and scientific discovery
- Recreation and tourism opportunities
- Cultural and spiritual enrichment

There are three main types of biodiversity:

1. Species diversity: The number of different species present in an ecosystem.
2. Genetic diversity: The variety of genes within a species or population.
3. Ecosystem diversity: The variety of ecosystems and habitats present in an area.

Overall, biodiversity is essential for maintaining the health of the planet and ensuring the well-being of all living things.

FLORA AND FAUNA

FLORA:

Flora is the plant life of a particular region or ecosystem, and it plays a vital role in sustaining life on Earth. Here are some interesting aspects of flora:

Types of Flora:

1. Trees: Oaks, pines, maples, and more
2. Shrubs: Roses, lilacs, and bushes
3. Flowers: Roses, daisies, sunflowers, and more
4. Grasses: Wheat, rice, corn, and lawn grass
5. Mosses and Lichens: Small, non-vascular plants
6. Ferns: Delicate, ancient plants
7. Fungi: Mushrooms, molds, and yeasts
8. Aquatic Plants: Algae, seaweed, and water lilies

Importance of Flora:

1. Oxygen Production: Plants produce oxygen through photosynthesis
2. Food Source: Fruits, vegetables, grains, and nuts
- 3.. Shelter and Habitat: Trees and plants provide shelter for animals
4. Soil Erosion Prevention: Roots hold soil in place
5. Climate Regulation: Plants absorb carbon dioxide and release oxygen
6. Medicine: Many plants have medicinal properties
7. Aesthetic Value: Beauty and fragrance of flowers and gardens

Flora is essential for human life, wildlife, and the health of the planet. It's important to conserve, appreciate, and protect the diverse range of plant species that inhabit our world.

FAUNA:

Fauna refers to the animals that inhabit a particular region or ecosystem. It encompasses a vast range of species, from tiny insects to massive mammals, and includes:

Types of Fauna:

1. Mammals: Warm-blooded animals like bears, wolves, deer, and humans
2. Birds: Winged creatures like eagles, robins, sparrows, and ostriches
3. Reptiles: Scaly animals like snakes, lizards, turtles, and crocodiles
4. Amphibians: Cold-blooded animals like frogs, toads, salamanders, and newts
5. Fish: Aquatic animals like salmon, sharks, goldfish, and guppies
6. Insects: Tiny creatures like bees, butterflies, ants, beetles, and grasshoppers
7. Arachnids: Eight-legged animals like spiders, scorpions, and ticks
8. Crustaceans: Shell-bearing animals like crabs, lobsters, shrimp, and crayfish

Importance of Fauna:

1. Food Source: Many animals are a source of food for humans and other animals
2. Ecosystem Balance: Fauna plays a crucial role in maintaining ecosystem balance
3. Pollination: Insects like bees and butterflies pollinate plants

4. Seed Dispersal: Animals help spread seeds, promoting plant growth
5. Decomposition: Animals like vultures and insects help break down organic matter
6. Medicine: Many animals are a source of medicinal products, like honey and venom
7. Conservation: Protecting fauna and their habitats is essential for maintaining biodiversity

Fauna is a vital component of the web of life, and its diversity and health are essential for maintaining the balance of ecosystems and the planet as a whole.

IMPORTANCE OF BIODIVERSITY

Biodiversity is essential for maintaining the health of the planet and ensuring the well-being of all living things. Some of the key importance of biodiversity includes:

1. Ecosystem Services: Biodiversity provides essential services like clean air and water, soil formation, and climate regulation.
2. Food Security: Many crops and livestock rely on biodiversity to pollinate, pest control, and maintain soil health.
3. Medicine: Biodiversity is a key source of new medicines and treatments, with many species still unknown to science.
4. Climate Change: Biodiversity helps regulate the climate by storing carbon, moderating weather patterns, and providing natural barriers to extreme weather events.
5. Recreation and Tourism: Biodiversity provides opportunities for outdoor recreation, tourism, and cultural enrichment.

6. Cultural Significance: Biodiversity has deep cultural and spiritual significance, with many species holding important cultural and spiritual values.

7. Resilience: Ecosystems with high biodiversity are more resilient to environmental changes and better able to adapt to new conditions.

8. Scientific Discovery: Biodiversity provides opportunities for scientific discovery and understanding of the natural world.

9. Support for Human Life: Biodiversity supports human life by providing essential resources like food, water, and shelter.

10. Intrinsic Value: Biodiversity has intrinsic value and is worth preserving for its own sake, regardless of its utility or functional value.

Overall, biodiversity is essential for maintaining the health of the planet and ensuring the well-being of all living things.

ECONOMIC VALUES-MEDICINAL PLANTS

Medicinal plants have significant economic values, including:

1. Pharmaceutical industry: Medicinal plants are used to develop new drugs and therapies, generating billions of dollars in revenue.

2. Traditional medicine: Herbal remedies and traditional medicine practices generate significant income, especially in developing countries.

3. Nutraceuticals and dietary supplements: Plant-based supplements and functional foods are a growing market, with increasing demand for natural health products.

4. Cosmetics and personal care: Plant-derived ingredients are used in skincare, haircare, and other personal care products, driving the cosmetics industry.

5. Agriculture and horticulture: Cultivation and trade of medicinal plants provide income opportunities for farmers and rural communities.
6. Ecotourism and sustainable tourism: Medicinal plant-based tourism, such as visiting botanical gardens or participating in herbal workshops, generates revenue and promotes conservation.
7. Research and development: Governments and companies invest in research on medicinal plants, driving innovation and economic growth.
8. Export and trade: Medicinal plants and their derivatives are traded globally, contributing to national economies and international trade.
9. Job creation: The medicinal plant industry employs people in various sectors, from cultivation to manufacturing and research.
10. Sustainable development: Medicinal plants can contribute to sustainable development goals, such as poverty reduction, healthcare access, and environmental conservation.

The economic value of medicinal plants is substantial, with the global market projected to reach \$5 trillion by 2025. However, it's important to ensure sustainable harvesting, fair trade practices, and conservation efforts to protect these valuable resources.

Medicinal Plants

Medicinal plants have been used for thousands of years to treat various ailments and maintain health. They contain naturally occurring compounds that can have therapeutic effects on the human body.

Here are some key points about medicinal plants:

Historical Significance: Many ancient civilizations, such as the Egyptians, Greeks, and Chinese, used medicinal plants. Traditional systems of medicine like Ayurveda and Traditional Chinese Medicine heavily rely on plant-based treatments.

Active Compounds: Medicinal plants contain bioactive compounds, such as alkaloids, flavonoids, tannins, and glycosides. These compounds can have anti-inflammatory, antioxidant, antimicrobial, and other therapeutic properties.

Common Medicinal Plants:

- **Aloe Vera:** Used for skin conditions, burns, and wounds.
- **Echinacea:** Known for boosting the immune system and fighting infections.
- **Garlic:** Has antimicrobial and cardiovascular benefits.
- **Ginger:** Effective for nausea, digestion, and anti-inflammatory purposes.
- **Turmeric:** Contains curcumin, which has anti-inflammatory and antioxidant properties.

Modern Medicine: Many modern pharmaceuticals are derived from medicinal plants. For example, aspirin was originally derived from willow bark, and the anti-cancer drug Taxol comes from the Pacific yew tree.

Safety and Efficacy: While medicinal plants can be beneficial, they must be used with caution. Dosage, preparation, and potential interactions with other medications need to be considered. Consulting healthcare professionals before using medicinal plants for therapeutic purposes is advisable.

Conservation: Overharvesting and habitat destruction pose threats to many medicinal plants. Sustainable practices and conservation efforts are crucial to preserving these valuable resources.

Economic values of Medicinal Plants:

Pharmaceutical Industry: Medicinal plants are a primary source of raw materials for the pharmaceutical industry. Many drugs are derived from or

inspired by compounds found in these plants. The global market for plant-derived drugs is substantial, contributing billions of dollars annually.

Traditional Medicine: In many countries, traditional medicine systems, such as Ayurveda, Traditional Chinese Medicine, and African traditional medicine, rely heavily on medicinal plants. These systems support a large number of practitioners and contribute significantly to the healthcare economy.

Agriculture and Horticulture: The cultivation of medicinal plants provides a source of income for farmers and agricultural workers. Specialized farming of these plants can be a lucrative business, particularly in regions where the climate and soil conditions are favorable.

Herbal Supplements: There is a growing market for herbal supplements and nutraceuticals, driven by consumer interest in natural health products. This market includes vitamins, dietary supplements, and herbal teas, which often use medicinal plant extracts.

Cosmetic Industry: Medicinal plants are also used in the cosmetic and personal care industry. Ingredients like aloe vera, chamomile, and lavender are commonly found in skincare and beauty products, enhancing their appeal and market value.

Research and Development: Medicinal plants are vital for scientific research, leading to the discovery of new drugs and therapies. Investments in R&D can lead to patents and commercialization opportunities, driving economic growth.

Employment Generation: The cultivation, processing, and commercialization of medicinal plants generate employment in rural and urban areas. This includes jobs in farming, harvesting, processing, marketing, and retailing.

Export Earnings: Many countries export medicinal plants and their derivatives, contributing to foreign exchange earnings. Countries like India and China are major exporters of medicinal plants and herbal products.

Sustainable Livelihoods: In many developing countries, medicinal plants provide a sustainable source of income for local communities, helping to alleviate poverty and support local economies.

Tourism: Ethnobotanical tourism, where visitors learn about medicinal plants and traditional healing practices, can also contribute to the economy, particularly in regions known for their rich biodiversity and traditional knowledge.

Drugs and Their Economic Values

Introduction

Drugs, encompassing both pharmaceutical products and illicit substances, have profound economic impacts globally. The pharmaceutical industry is a cornerstone of modern healthcare systems, contributing significantly to global GDP, driving innovation, and improving public health. Conversely, the illicit drug trade has substantial social and economic costs, affecting economies through healthcare costs, crime, and lost productivity. This document delves into the economic values of both pharmaceutical and illicit drugs, providing a comprehensive overview of their impacts.

The Pharmaceutical Industry

Market Overview

The global pharmaceutical market is one of the most critical sectors in the global economy. As of 2023, it was valued at approximately \$1.4 trillion and is projected to reach \$1.8 trillion by 2026, growing at a compound annual growth rate (CAGR) of 5-6%. This growth is driven by factors such as an aging population, increasing prevalence of chronic diseases, and advancements in biotechnology and pharmaceuticals.

Prescription Drugs

Prescription drugs represent the largest segment of the pharmaceutical market. These medications are prescribed by healthcare professionals and are used to treat various conditions, including chronic diseases such as diabetes, cardiovascular diseases, and cancer. The demand for prescription drugs is growing due to the increasing prevalence of these diseases and advancements in medical treatments. Major players in this segment include Pfizer, Novartis, Roche, and Merck.

Over-the-Counter (OTC) Drugs

OTC drugs are medications available without a prescription and are used for common ailments such as colds, allergies, and minor pain relief. The OTC market is also significant, with leading manufacturers including Johnson & Johnson, Bayer, and GlaxoSmithKline. The availability and accessibility of OTC drugs provide consumers with convenient treatment options and reduce the burden on healthcare systems.

Biopharmaceuticals

Biopharmaceuticals, including biologics, biosimilars, and gene therapies, represent the fastest-growing segment of the pharmaceutical industry. These advanced therapies are developed through biotechnology and offer targeted treatments for various diseases, including rare and complex conditions. Key companies in this segment include Amgen, Biogen, and Gilead Sciences. The growth of biopharmaceuticals is driven by continuous advancements in biotechnology and a growing understanding of genetic and molecular mechanisms of diseases.

Economic Impact

Employment

The pharmaceutical industry is a major employer worldwide, providing jobs in research and development (R&D), manufacturing, marketing, and distribution. In the United States alone, the industry employs over 800,000 people, with millions more employed indirectly in related sectors. These jobs contribute significantly to local and national economies, supporting economic growth and stability.

Research and Development (R&D)

R&D is a cornerstone of the pharmaceutical industry, with companies investing heavily in the development of new drugs and therapies. Annual R&D expenditures exceed \$200 billion globally, driving innovation and the discovery of breakthrough treatments. This investment not only leads to new medications but also supports a wide range of scientific research and technological advancements.

Healthcare Costs

Pharmaceuticals contribute significantly to healthcare costs, accounting for a substantial portion of healthcare expenditures. However, effective medications can reduce overall healthcare costs by preventing or managing diseases, thereby reducing the need for more expensive treatments and hospitalizations. For example, effective management of chronic diseases with medication can prevent complications and hospital admissions, ultimately saving healthcare systems money.

The Illicit Drug Trade

The illicit drug trade is a vast and complex global industry, with annual revenues estimated to be between \$426 billion and \$652 billion according to the United Nations Office on Drugs and Crime (UNODC). This trade involves the production, trafficking, and sale of illegal substances, impacting countries worldwide. The dynamics of the illicit drug trade are influenced by various factors, including demand, supply chains, and law enforcement efforts.

Cannabis

Cannabis is the most widely consumed illicit drug globally. However, the legal landscape is changing, with several countries and states legalizing its use for medical and recreational purposes. This shift has created a burgeoning legal market, particularly in the United States, where the cannabis industry is valued at over \$20 billion annually. The legalization of cannabis has economic benefits, including tax revenues, job creation, and reduced law enforcement costs. However, it also poses challenges, such as regulation and ensuring public safety.

Cocaine

Cocaine is primarily produced in South America, with Colombia being the largest producer. The drug is trafficked globally, generating billions in revenue for drug cartels. The cocaine trade has significant economic impacts, including the costs associated with law enforcement, healthcare, and the social consequences of addiction and violence. Efforts to combat cocaine production and trafficking involve substantial expenditures by governments and international organizations.

Opioids

The opioid crisis, particularly in North America, has had devastating economic and social impacts. Opioids, including heroin and synthetic opioids like fentanyl, are responsible for a significant number of overdose deaths. The crisis has strained healthcare systems, increased law enforcement costs, and led to lost productivity. Addressing the opioid epidemic requires comprehensive strategies, including prevention, treatment, and policy interventions.

Amphetamines and Other Stimulants

Amphetamines, including methamphetamine and MDMA (ecstasy), are widely used stimulants with significant economic impacts. The production and trafficking of these drugs contribute to criminal activity and associated costs.

Additionally, the health consequences of stimulant abuse place a burden on healthcare systems. Efforts to combat stimulant abuse include law enforcement, public health initiatives, and international cooperation.

Fisheries

Introduction to Fisheries

Fisheries are the activities related to catching, processing, and selling fish and other aquatic organisms.

They play a crucial role in providing food, livelihoods, and economic opportunities for millions of people worldwide.

Sustainable management of fisheries is essential to ensure the health of marine ecosystems and the long-term viability of fish stocks.

Types of Fisheries

1. Commercial Fisheries: Focus on catching fish and other aquatic species for sale and profit. These can range from small scale artisanal fisheries to large industrial operations.

2. Recreational Fisheries: Involve fishing for sport or leisure rather than for commercial profit. This sector can also contribute significantly to local economies through tourism. fishing primarily for personal or local community consumption, often crucial for food security in coastal and rural communities.

3. Aquaculture: Also known as fish farming, aquaculture is the cultivation of aquatic organisms such as fish, crustaceans, mollusks, and aquatic plants under controlled conditions. Aquaculture is one of the fastest-growing sectors in food production, helping to meet the increasing demand for seafood.

4. Economic Importance:- Fisheries contribute significantly to the global economy, providing jobs and income for millions of people.
- They are a major source of export revenue for many countries, especially those with extensive coastlines. -

The sector supports various related industries, including boat manufacturing, gear production ,and seafood processing.

5. Food Security and Nutrition:

- Fish and other seafood are vital sources of protein and essential nutrients for billions of people worldwide.

- They play a crucial role in the diet of many coastal and island populations.

6. Biodiversity and Ecosystem Services:

- Healthy fisheries are integral to maintaining marine biodiversity and ecosystem health.
- They provide ecosystem services such as nutrient cycling, habitat provision, and carbon sequestration.

7. Sustainability and Management:

- Overfishing, habitat destruction, pollution, and climate change pose significant threats to fisheries.
- Sustainable fisheries management practices are essential to ensure the long-term viability of fish stocks and marine ecosystems. This includes regulations like catch limits, protected areas, and measures to reduce bycatch (unintended catch of non-target species).

8. Technological Advancements:

Advances in fishing technology and aquaculture practices have increased efficiency and productivity.

Innovations in monitoring and surveillance help in the management and conservation of fish stocks.

9. Global Organizations and Agreements:

- Organizations such as the Food and Agriculture Organization (FAO) of the United Nations play a key role in promoting sustainable fisheries.
- International agreements and regional fisheries management organizations (RFMOs) work to regulate fishing activities and promote conservation.

Importance of Fisheries

- Fisheries are a vital source of protein and essential nutrients for billions of people around the world, especially in developing countries.
- They contribute significantly to global food security and provide employment for millions of people in the fishing industry.
- Fisheries also support related industries such as seafood processing, transportation, and marketing, driving economic growth in coastal communities.

Challenges Facing Fisheries

1. Overfishing, illegal fishing, and bycatch are major threats to the sustainability of fisheries and marine ecosystems.
2. Climate change, pollution, habitat destruction, and ocean acidification are exacerbating the pressures on fish stocks and marine biodiversity.
3. Poor governance, lack of enforcement, and inadequate monitoring and control measures further undermine efforts to manage fisheries sustainably.

Sustainable Fisheries Management

Sustainable fisheries management involves setting catch limits, implementing conservation measures, and monitoring fish populations to prevent overexploitation.

Ecosystem-based approaches consider the interactions between fish, marine habitats, and other species to ensure the health of the entire ecosystem.

Stakeholder engagement, scientific research, and adaptive management strategies are essential components of effective fisheries management.

Technological Innovations in Fisheries

Advances in technology, such as GPS tracking, sonar systems, and satellite imagery, have improved the efficiency and sustainability of fishing operations.

Remote sensing technologies can help monitor fishing activities, detect illegal fishing, and enforce regulations in marine protected areas.

Electronic monitoring and surveillance tools enable real-time data collection and analysis to support evidence-based decision-making in fisheries management.

International Cooperation in Fisheries

Regional fisheries management organizations (RFMOs) play a crucial role in coordinating efforts to manage shared fish stocks in international waters.

The United Nations Convention on the Law of the Sea (UNCLOS) provides a legal framework for regulating fisheries activities and protecting marine biodiversity.

International agreements, such as the Port State Measures Agreement and the FAO Code of Conduct for Responsible Fisheries, promote sustainable fisheries practices and combat illegal fishing globally.

Future Trends in Fisheries

The future of fisheries will be shaped by emerging challenges such as climate change, globalization, and evolving consumer preferences for sustainable seafood.

Innovations in aquaculture, seafood traceability, and marine conservation technologies are expected to drive positive change in the fishing industry.

Collaboration between governments, industry, NGOs, and local communities will be crucial to ensure the long-term sustainability and resilience of fisheries in a rapidly changing world.

Introduction to Biogeochemical Cycles:**Definition :**

Biogeochemical cycles describe the movement of elements and compounds through living organisms (bio-), the earth (geo-), and chemical processes (-chemical). They are natural pathways by which essential elements of living matter circulate through the biosphere, atmosphere, hydrosphere, and lithosphere.

The earth obtains energy from the sun which is radiated back as heat, rest all other elements are present in a closed system. The major elements include:

- Carbon
- Hydrogen
- Nitrogen
- Oxygen
- Phosphorus
- Sulphur

These elements are recycled through the biotic and abiotic components of the ecosystem. The atmosphere, hydrosphere and lithosphere are the abiotic components of the ecosystem.

Types of Biogeochemical Cycles

Biogeochemical cycles are basically divided into two types:

- **Gaseous cycles** – Includes Carbon, Oxygen, Nitrogen, and the Water cycle.
- **Sedimentary cycles** – Includes Sulphur, Phosphorus, Rock cycle, etc.

Let us have a look at each of these biogeochemical cycles in brief:

Water cycle :

- The water from the different water bodies evaporates, cools, condenses and falls back to the earth as rain.
- This biogeochemical cycle is responsible for maintaining weather conditions. The water in its various forms interacts with the surroundings and changes the temperature and pressure of the atmosphere.
- There's another process called Evapotranspiration (i.e. vapour produced from leaves) which aids this process. It is the evaporation of water from the leaves, soil and water bodies to the atmosphere which again condenses and falls as rain.

Carbon cycle :

- It is one of the biogeochemical cycles in which carbon is exchanged among the biosphere, geosphere, hydrosphere, atmosphere and pedosphere.
- All green plants use carbon dioxide and sunlight for photosynthesis. Carbon is thus stored in the plant. The green plants, when dead, are buried into the soil that gets converted into fossil fuels made from carbon. These fossil fuels when burnt, release carbon dioxide into the atmosphere.
- Also, the animals that consume plants, obtain the carbon stored in the plants. This carbon is returned to the atmosphere when these animals decompose after death. The carbon also returns to the environment through cellular respiration by animals.
- Huge carbon content in the form of carbon dioxide is produced that is stored in the form of fossil fuel (coal & oil) and can be extracted for various commercial and non-commercial purposes. When factories use these fuels, the carbon is again released back in the atmosphere during combustion.

Nitrogen cycle :

- It is the biogeochemical cycle by which nitrogen is converted into several forms and it gets circulated through the atmosphere and various ecosystems such as terrestrial and marine ecosystems.

- Nitrogen is an essential element of life. The nitrogen in the atmosphere is fixed by the nitrogen-fixing bacteria present in the root nodules of the leguminous plants and made available to the soil and plants.
- The bacteria present in the roots of the plants convert this nitrogen gas into a usable compound called ammonia. Ammonia is also supplied to plants in the form of fertilizers. This ammonia is converted into nitrites and nitrates. The denitrifying bacteria reduce the nitrates into nitrogen and return it into the atmosphere.

Oxygen cycle :

- This biogeochemical cycle moves through the atmosphere, the lithosphere and the biosphere. Oxygen is an abundant element on our Earth. It is found in the elemental form in the atmosphere to the extent of 21%.
- Oxygen is released by the plants during photosynthesis. Humans and other animals inhale the oxygen exhale carbon dioxide which is again taken up by the plants. They utilise this carbon dioxide in photosynthesis to produce oxygen, and the cycle continues.

Phosphorus Cycle :

In this biogeochemical cycle, phosphorus moves through the hydrosphere, lithosphere and biosphere. Phosphorus is extracted by the weathering of rocks. Due to rains and erosion phosphorus is washed away in the soil and water bodies. Plants and animals obtain this phosphorus through the soil and water and grow. Microorganisms also require phosphorus for their growth. When the plants and animals die they decompose, and the stored phosphorus is returned to the soil and water bodies which is again consumed by plants and animals and the cycle continues.

Sulphur Cycle:

This biogeochemical cycle moves through the rocks, water bodies and living systems. Sulphur is released into the atmosphere by the weathering of rocks and is converted into sulphates. These sulphates are taken up by the microorganisms and plants and converted into organic forms. Organic sulphur is consumed by animals through food. When the animals die and decompose, sulphur is returned

to the soil, which is again obtained by the plants and microbes, and the cycle continues.

Importance:

- These cycles ensure the sustainability of ecosystems by recycling essential elements, supporting life processes, and maintaining ecological balance.
- They ensure the continuous supply of essential nutrients needed for plant growth, which supports agriculture and natural vegetation.
- Facilitate the flow of energy through ecosystems via the carbon cycle, including photosynthesis and respiration
- Regulate climate by controlling the concentration of greenhouse gases like carbon dioxide and methane
- Biogeochemical cycles help in the regulation of natural elements that are necessary for living beings, by channelling through physical and biological phenomena. It acts as a recycling procedure in nature

Major Threats to Biodiversity:

- ❖ Climate change
- ❖ Pollution
- ❖ Overexploitation of species
- ❖ Habitat loss
- ❖ Invasive Species

Climate change:

climate change is also a consequence of human population needs for energy, and the use of fossil fuels to meet those needs. Essentially, burning fossil fuels, including as oil, natural gas, and coal, increases carbon dioxide concentrations in the atmosphere. Carbon dioxide traps heat energy from the sun, resulting not only in an average increase in global temperature but also in changing precipitation patterns and increased frequency and severity of extreme weather events, such as hurricanes. Scientists overwhelmingly agree the present warming trend is caused by humans. For example, pollinating insects typically emerge in the spring based on temperature cues.

Pollution:

Pollution occurs when chemicals, particles, or other materials are released into the environment, harming the organisms there. For many thousands of years, ever since they built the first campfire, human activity has generated air, water, and soil pollution. For most of human history, however, these contaminants had relatively little environmental impact. But over the last few centuries, pollution levels skyrocketed as a result of population growth and the Industrial Revolution. As a result, regulations have been enacted to control emissions. Pollution has contributed to the decline of many threatened species. For example, a 2007 study by Kingsford and colleagues found that pollution was a major pressure on 30% of threatened Red List species in Australia and surrounding regions.

Overexploitation of species:

Overexploitation (overharvesting) involves hunting, fishing, or otherwise collecting organisms at a faster rate than they can be replenished. Terrestrial animals may be overexploited as sources of food, garments, medicine, or pets. For example, the poaching of elephants for their valuable ivory and rhinos for their horns, which are used in traditional medicine, is a major threat to these species. There are also concerns about the effect of the pet trade on some terrestrial species such as turtles, amphibians, birds, plants, and even the orangutans. Harvesting of pangolins for their scales and meat, and as curiosities, has led to a drastic decline in population size.

Habitat loss:

Humans rely on technology to modify their environment and make it habitable. Other species cannot do this. Elimination of their habitat—whether it is a forest, coral reef, grassland, or flowing river—will kill the individuals in the species. Remove the entire habitat, and the species will become extinct, unless they are among the few species that do well in human-built environments. Habitat loss includes habitat destruction and habitat fragmentation.

Invasive Species:

Invasive species are non-native organisms that, when introduced to an area out of its native range, disrupt the community they invade. Non-native (exotic) refers to species occurring outside of their historic distribution. Invasive species have been intentionally or unintentionally introduced by humans into an ecosystem in which they did not evolve. Human transportation of people and goods, including

the intentional transport of organisms for trade, has dramatically increased the introduction of species into new ecosystems. In the United States, invasive species like the purple loosestrife (*Lythrum salicaria*) and the zebra mussel (*Dreissena polymorpha*) have drastically altered the ecosystems they invaded.

Natural and Anthropogenic Disturbances to Biodiversity

Introduction

Biodiversity, the variety of life on Earth, is essential for ecosystem health and stability. It provides numerous ecosystem services that are vital for human survival, such as clean air and water, pollination of crops, and regulation of climate. However, biodiversity is threatened by both natural and anthropogenic disturbances. Understanding these disturbances and their impacts is crucial for developing effective conservation strategies. This document explores the types of disturbances affecting biodiversity, their causes, and their consequences.

Natural Disturbances to Biodiversity

Types of Natural Disturbances

Natural disturbances are events that disrupt ecosystems, communities, or populations and cause changes in the structure and function of ecosystems. These disturbances can be episodic or periodic and include events such as:

1. Wildfires:

- Natural wildfires are often ignited by lightning strikes.
- They play a crucial role in maintaining the health of certain ecosystems by clearing out dead vegetation, promoting new growth, and maintaining species diversity.

2. Floods:

- Flooding can reshape landscapes, create new habitats, and replenish nutrients in the soil.
- Periodic flooding is essential for the life cycles of many species, especially in floodplain and wetland ecosystems.

3. Hurricanes and Typhoons:

- These powerful storms can cause significant destruction to coastal and inland ecosystems.

- They can also create new habitats and opportunities for colonization by different species.

4. Volcanic Eruptions:

- Volcanic activity can destroy existing habitats but also create new ones as ecosystems recover and new species colonize the area.
- Volcanic ash can enrich soils with minerals, promoting plant growth.

5. Droughts:

- Extended periods of drought can stress ecosystems and lead to shifts in species composition.
- Some species and ecosystems are adapted to withstand and recover from periodic droughts.

Impacts of Natural Disturbances

Natural disturbances often have complex and variable impacts on biodiversity. Some of the effects include:

- **Enhanced Diversity:** Many natural disturbances create a mosaic of habitats at different stages of succession, promoting species diversity and ecosystem resilience.
- **Adaptation and Evolution:** Species that survive natural disturbances often have adaptations that allow them to cope with change, contributing to evolutionary processes.
- **Regeneration and Renewal:** Natural disturbances can lead to the regeneration of ecosystems, cycling nutrients, and promoting the growth of new vegetation.

Anthropogenic Disturbances to Biodiversity

Types of Anthropogenic Disturbances

Anthropogenic, or human-induced, disturbances are activities that directly or indirectly affect ecosystems and biodiversity. These disturbances are often more frequent and intense than natural ones and include:

1. Habitat Destruction:

- Activities such as deforestation, urbanization, and agriculture lead to the loss and fragmentation of habitats.
- Habitat destruction is one of the leading causes of biodiversity loss globally.

2. Pollution:

- Pollution from industrial, agricultural, and urban sources contaminates air, water, and soil.
- Chemicals, plastics, and other pollutants can have toxic effects on wildlife and ecosystems.

3. Climate Change:

- Human activities, particularly the burning of fossil fuels, release greenhouse gases that drive climate change.
- Climate change affects temperature and precipitation patterns, sea levels, and the frequency and intensity of extreme weather events.

4. Overexploitation:

- Overfishing, hunting, and harvesting of plants and animals can deplete populations and lead to extinction.
- Sustainable management practices are essential to prevent overexploitation.

5. Invasive Species:

- Human activities often introduce non-native species to new environments, where they can become invasive.
- Invasive species can outcompete, prey on, or bring diseases to native species, disrupting ecosystems.

Impacts of Anthropogenic Disturbances

Anthropogenic disturbances have profound and often detrimental effects on biodiversity:

- **Habitat Loss and Fragmentation:** The destruction and fragmentation of habitats reduce the available space for species, leading to population declines and extinctions.

- **Pollution and Contamination:** Pollutants can have lethal or sub-lethal effects on species, disrupt reproductive and developmental processes, and degrade habitats.
- **Climate Change Effects:** Changes in climate can alter habitats, force species to migrate, and increase the vulnerability of ecosystems to other disturbances.
- **Overexploitation Consequences:** Unsustainable use of resources can deplete populations, disrupt food webs, and lead to the collapse of ecosystems.
- **Impact of Invasive Species:** Invasive species can significantly alter ecosystems, reduce native biodiversity, and cause economic damage.

Zoological Garden

A Zoological garden, commonly known as a zoo, is a facility where animals are housed within enclosures, displayed to the public, and often bred for conservation purposes. Zoos aim to educate visitors about wildlife and conservation efforts, while also providing research opportunities to study animal behavior, biology, and ecology. They vary widely in size and scope, from small local zoos to large institutions with extensive collections of diverse species. Modern zoos also prioritize animal welfare, enrichment programs, and efforts to conserve endangered species through breeding programs and habitat preservation initiatives.

Here are some key aspects and functions of modern zoological gardens:

- ❖ **Conservation:** Many zoos actively participate in conservation efforts, particularly through breeding programs of endangered species. These programs aim to maintain genetic diversity and prevent extinction.
- ❖ **Education:** Zoos play a crucial role in educating the public about wildlife, biodiversity, and conservation issues. They often provide interpretive signage, guided tours, and educational programs for visitors of all ages.
- ❖ **Research:** Zoos contribute to scientific research in various fields such as animal behaviors, nutrition, reproduction, and veterinary medicine. Studying animals in controlled environments can provide insights that are difficult to obtain in the wild.
- ❖ **Animal Welfare:** Modern zoos prioritize the welfare of their animals. This includes providing appropriate enclosures that mimic natural habitats, enrichment activities to stimulate natural behaviors, and veterinary care to ensure health and well-being.
- ❖ **Public Engagement:** Zoos serve as popular recreational destinations where families and individuals can observe and learn about a wide variety of animals from around the world. They often host events, exhibitions, and interactive experiences to engage visitors.
- ❖ **Collaboration:** Zoos frequently collaborate with each other, as well as with conservation organizations, universities, and governmental agencies, to promote global conservation initiatives and share knowledge and resources.
- ❖ **Species Preservation:** Some zoos participate in species reintroduction programs, where animals bred in captivity are released into protected habitats in the wild to bolster wild populations or reestablish them in areas where they have become locally extinct.
- ❖ **Ethical Considerations:** Zoos are increasingly addressing ethical considerations, such as the impact of captivity on animal welfare, the importance of natural behavior expression, and the ethics of breeding programs and animal display.
- ❖ **Community Outreach:** Many zoos engage with local communities through outreach programs, partnerships with schools, and collaborations with conservation organizations to raise awareness about environmental issues and promote sustainable practices.

Overall, while controversies exist regarding the ethical and moral implications of keeping animals in captivity, modern zoological gardens continue to evolve, aiming to balance conservation, education, research, and ethical considerations in their missions.

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Purpose and Functions:

1. **Conservation:** Many zoos are actively involved in conservation efforts, including breeding endangered species, conducting research on genetics and behavior, and participating in species reintroduction programs.
2. **Education:** Zoos aim to educate the public about animals, their habitats, and conservation issues. They often provide interpretive exhibits, educational programs, guided tours, and outreach initiatives for schools and community groups.
3. **Research:** Zoos conduct scientific research to enhance animal welfare, improve breeding programs, study animal behavior, and contribute to broader conservation biology and veterinary medicine.
4. **Recreation:** Zoos provide recreational opportunities for visitors to observe and learn about animals up close, fostering appreciation for wildlife and biodiversity.
5. **Animal Welfare:** Modern zoos prioritize animal welfare through appropriate enclosure design, enrichment programs, veterinary care, and nutrition management.

Types of Zoos:

- **Traditional Zoos:** These zoos exhibit a wide range of animals in naturalistic enclosures.
- **Aquariums:** Focus on aquatic animals, showcasing marine and freshwater species.
- **Aviaries:** Specialize in birds, providing large enclosed habitats for aviary species.
- **Wildlife Parks:** Often larger than traditional zoos, with larger enclosures resembling natural habitats for free-roaming animals.

Ethics and Controversies:

- **Animal Welfare:** Concerns about captivity, stress, and psychological well-being of animals in zoos.
- **Conservation vs. Entertainment:** Debate over the balance between entertainment value and conservation/educational goals.
- **Breeding Programs:** Criticism and support regarding captive breeding efforts for endangered species.

Modern Trends:

- **Conservation Initiatives:** Increasing focus on conservation breeding, habitat restoration, and species recovery programs.
- **Education and Outreach:** Enhancing educational programs and digital engagement to reach wider audiences.
- **Visitor Experience:** Improving guest experience through immersive exhibits, interactive displays, and conservation-themed attractions.

Regulations and Standards:

- **Accreditation:** Many zoos seek accreditation from organizations like the Association of Zoos and Aquariums (AZA) or the European Association of Zoos and Aquaria (EAZA), which set standards for animal care, conservation, and visitor safety.

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- **Government Regulations:** Zoos are subject to national and local regulations regarding animal welfare, public safety, and conservation practices.

Conclusion:

Zoological gardens play a multifaceted role in society, balancing conservation, education, research, and public enjoyment. While they face challenges and ethical considerations, responsible zoos contribute significantly to wildlife conservation efforts and public understanding concepts.

Gene Banks:

Gene banks are facilities that conserve genetic material from plants, animals, or microorganisms. They play a crucial role in preserving biodiversity, supporting agricultural sustainability, and ensuring food security. Here's an overview of gene banks:

Types of Gene Banks:

1. **Seed Banks:** Store seeds under controlled conditions to maintain their viability. Examples include the Svalbard Global Seed Vault and the Millennium Seed Bank.
2. **Field Gene Banks:** Live plants are grown and maintained in fields or gardens. This method is often used for species with seeds that do not survive drying and freezing well.
3. **In Vitro Gene Banks:** Preserve plant tissues or organs (like shoot tips or embryos) in sterile, controlled environments. This method is useful for plants that are difficult to store as seeds.
4. **Cryobanks:** Use liquid nitrogen to store genetic material at ultra-low temperatures. This is a common method for preserving animal genetic material such as sperm, eggs, or embryos.
5. **DNA Banks:** Store purified DNA samples, which can be used for genetic research and breeding programs.

Importance of Gene Banks:

1. **Biodiversity Conservation:** Gene banks help protect species from extinction, especially those threatened by habitat loss, climate change, or human activities.
2. **Agricultural Sustainability:** Preserve a wide variety of crop species and their wild relatives, which can be crucial for breeding programs aimed at improving crop resilience to pests, diseases, and changing climate conditions.
3. **Research and Development:** Provide genetic material for scientific research, helping to understand genetic diversity, evolution, and the genetic basis of traits.
4. **Food Security:** Ensure the availability of genetic resources that can be used to develop new, more productive, and resilient crop varieties, thereby supporting global food supply.

Techniques for Plant Gene Banks

1. **Seed Storage:**

Biodiversity

- **Dry Seed Storage:** Seeds are dried to a low moisture content and stored at low temperatures, often in sealed containers. This is the most common method for orthodox seeds (seeds that tolerate drying and freezing).
 - **Cryopreservation:** Seeds or embryonic tissues are stored at ultra-low temperatures in liquid nitrogen (-196°C). This technique is used for recalcitrant seeds (seeds that do not tolerate drying) and other plant tissues.
 - **Vacuum Packaging:** Seeds are stored in vacuum-sealed containers to reduce moisture and oxygen levels, prolonging their viability.
2. **In Vitro Storage:**
- **Tissue Culture:** Plant tissues (e.g., shoot tips, embryos) are cultured in a sterile, controlled environment on nutrient media. This method is used for species that produce recalcitrant seeds or have vegetative propagation.
 - **Slow Growth Storage:** Plant tissues are cultured at reduced temperatures and with modified growth media to slow down their growth, extending the storage period without frequent subculturing.
3. **Field Gene Banks:**
- **Living Collections:** Plants are grown in fields or gardens and maintained as living collections. This method is used for perennial species, vegetatively propagated plants, and species with recalcitrant seeds.

Techniques for Animal Gene Banks

1. **Cryopreservation of Gametes and Embryos:**
- **Sperm Cryopreservation:** Sperm cells are collected, processed, and frozen in liquid nitrogen. This technique is widely used for livestock, poultry, and endangered species.
 - **Oocyte and Embryo Cryopreservation:** Oocytes (eggs) and embryos are collected and frozen. This method is essential for preserving female genetic material and for use in assisted reproductive technologies.
2. **Somatic Cell Cryopreservation:**
- **Fibroblast and Other Somatic Cells:** Cells are collected from tissues (e.g., skin biopsies) and cryopreserved. These cells can be used for cloning and genetic studies.
3. **DNA and Genetic Material Storage:**
- **Genomic DNA:** DNA extracted from tissues or blood samples is stored for genetic analysis and research.
 - **Cryopreserved Tissues:** Tissues are collected and stored at ultra-low temperatures for future DNA extraction and research.

Techniques for Microbial Gene Banks

1. **Cryopreservation:**
- Microbial cultures are frozen in liquid nitrogen or at -80°C for long-term storage. This method is widely used for bacteria, fungi, and other microorganisms.
2. **Lyophilization (Freeze-Drying):**
- Microbial cultures are frozen and then dehydrated under a vacuum. The resulting powder can be stored at room temperature and rehydrated when needed.
3. **Agar Slant Cultures:**

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- Microorganisms are cultured on agar slants and stored at low temperatures. This technique is commonly used for short to medium-term storage.

Techniques for Fish and Aquatic Organisms

1. **Cryopreservation of Sperm:**
 - Sperm from fish and other aquatic organisms are cryopreserved for future use in breeding and conservation programs.
2. **Cryopreservation of Embryos and Larvae:**
 - Embryos and larvae are frozen and stored at ultra-low temperatures, although this technique is more challenging and less common than sperm cryopreservation.

Techniques for Invertebrate Gene Banks

1. **Cryopreservation of Eggs and Sperm:**
 - Eggs and sperm from invertebrates, such as insects and mollusks, are cryopreserved for future use in breeding and research.
2. **Preservation of Living Cultures:**
 - Invertebrates are maintained as living cultures in controlled environments, such as insectaries or aquariums, for species that are difficult to cryopreserve.

General Considerations

- **Viability Testing:** Regular testing of stored genetic material is essential to ensure its viability and genetic integrity over time.
- **Documentation and Data Management:** Comprehensive records and databases are maintained to track the genetic material, including information on collection, storage conditions, and usage history.
- **Biosecurity and Safety:** Measures are in place to prevent contamination, loss, or unauthorized access to the genetic material.

Notable Gene Banks:

1. **Svalbard Global Seed Vault (Norway):** Often referred to as the "Doomsday Vault," it stores duplicates of seed samples from gene banks worldwide.
2. **Millennium Seed Bank (UK):** The largest and most diverse wild plant seed bank in the world, focused on conserving seeds from across the globe.
3. **National Center for Genetic Resources Preservation (USA):** A major facility that preserves a wide range of genetic materials, including seeds, tissues, and DNA.

Challenges and Future Directions:

1. **Funding and Resources:** Maintaining gene banks requires significant financial and technical resources.
2. **Technological Advances:** Continued development of preservation techniques and technologies to improve the efficiency and effectiveness of gene banks.

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3. **Accessibility and Sharing:** Ensuring that genetic resources are accessible to researchers and breeders worldwide, while respecting the rights of countries and communities that contribute genetic material.
4. **Climate Change:** Adapting gene bank practices to ensure the long-term viability of stored genetic material in the face of changing environmental conditions.

Gene banks are vital for preserving the genetic diversity necessary for the sustainability and resilience of ecosystems and agricultural systems worldwide.

Pollen Culture:

Pollen culture, also known as microspore culture, is a plant breeding technique used to produce haploid plants from pollen grains. Here's a detailed overview:

Pollen culture involves the in vitro culture of pollen grains (microspores) to develop into haploid plants. These haploid plants have only one set of chromosomes (n), compared to the diploid state (2n) of normal plants.

Types of pollen culture:

- **Anther Culture:** This method involves culturing the whole anther containing the pollen grains. It is relatively straightforward but can sometimes result in a mixture of diploid and haploid plants due to the presence of somatic tissues.
- **Microspore Culture:** In this method, isolated microspores (pollen grains) are cultured directly. This technique is more precise and can lead to a higher proportion of haploid plants. It's more technically demanding but useful for specific research and breeding applications.
- **Pollen Suspension Culture:** This involves suspending pollen grains in a liquid medium. It is less commonly used due to the difficulty in maintaining and manipulating the cultures, but it can be beneficial for certain experimental conditions.
- **Immature Pollen Culture:** Culturing pollen at an early stage of development can sometimes lead to better success rates in haploid production. This method requires precise timing and control over the developmental stage of the pollen.
- **Androgenesis:** This is a broader term encompassing both anther and microspore culture techniques aimed at producing haploid plants through the development of male gametophytes.

Steps in Pollen Culture:

1. **Selection of Donor Plant:** The donor plant should be healthy and in the right stage of development. Generally, the anthers containing the microspores are collected from the flower buds before they dehisce.

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2. **Sterilization:** The collected anthers are sterilized to prevent contamination. This can involve washing with disinfectants like ethanol or bleach, followed by rinsing with sterile water.
3. **Isolation of Microspores:** Microspores are isolated from the anthers through crushing and filtration techniques.
4. **Culture Medium:** The isolated microspores are placed in a suitable culture medium. This medium typically contains nutrients, growth regulators, and sometimes hormones to support the growth and development of microspores.
5. **Incubation:** The cultured microspores are incubated under controlled environmental conditions such as temperature, light, and humidity. The conditions are optimized to induce the development of microspores into embryos and subsequently into haploid plants.
6. **Regeneration:** The haploid embryos are transferred to a regeneration medium where they develop into complete haploid plants.
7. **Doubling of Chromosomes:** Haploid plants are often treated with chemicals like colchicine to double their chromosome number, producing homozygous diploid plants. This step is crucial for plant breeding as it results in uniform and stable genetic lines.

Applications of Pollen Culture:

1. **Plant Breeding:** Pollen culture is used to produce pure lines quickly, which are essential for hybrid seed production and other breeding programs.
2. **Genetic Studies:** Haploid plants are useful in genetic studies as they express recessive traits, making it easier to study gene function and inheritance patterns.
3. **Crop Improvement:** This technique is used to develop disease-resistant, high-yielding, and stress-tolerant crop varieties.

Advantages:

- **Speed:** It accelerates the breeding process by producing homozygous lines in one generation.
- **Efficiency:** It allows for the rapid screening of genetic traits.
- **Precision:** Enables the production of genetically uniform plants.

Limitations:

- **Species Specificity:** Not all plant species respond well to pollen culture.
- **Technical Challenges:** Requires specialized equipment and expertise.
- **Low Efficiency:** Sometimes, the success rate can be low depending on the species and conditions.

Pollen culture is a valuable tool in modern plant biotechnology, contributing significantly to agricultural advancements and food security.

Ecological Restoration

Ecological restoration is the process of assisting the recovery of an ecosystem that has been degraded, damaged, or destroyed. The goal is to return the ecosystem to its original structure,

function, and diversity, or to create a new, resilient system that can provide essential ecosystem services. Here are the key components and approaches involved in ecological restoration:

Key Components of Ecological Restoration

1. **Assessment and Planning:**
 - **Baseline Assessment:** Understanding the current state of the ecosystem, including the extent of degradation, existing biodiversity, soil conditions, and water quality.
 - **Goal Setting:** Defining clear, achievable objectives for the restoration project based on ecological, social, and economic factors.
 - **Stakeholder Involvement:** Engaging local communities, landowners, scientists, and policymakers in the planning process.
2. **Restoration Techniques:**
 - **Revegetation:** Planting native species to restore plant communities. This can include seeding, planting nursery-grown plants, or facilitating natural regeneration.
 - **Invasive Species Management:** Removing or controlling invasive species that outcompete native flora and fauna.
 - **Erosion Control:** Implementing measures to prevent soil erosion, such as planting cover crops, installing silt fences, or building terraces.
 - **Hydrological Restoration:** Restoring natural water flow patterns by removing barriers, re-establishing wetlands, or creating buffer zones along water bodies.
 - **Soil Amendments:** Adding organic matter, nutrients, or beneficial microorganisms to improve soil health and fertility.
3. **Monitoring and Maintenance:**
 - **Regular Monitoring:** Tracking the progress of the restoration efforts through regular surveys, data collection, and analysis.
 - **Adaptive Management:** Adjusting strategies and techniques based on monitoring results to ensure the project stays on track.
 - **Long-term Maintenance:** Ensuring the restored ecosystem remains healthy and resilient through ongoing management and protection.
4. **Education and Outreach:**
 - **Community Engagement:** Involving local communities in restoration activities and educating them about the importance of ecosystem health.
 - **Public Awareness Campaigns:** Raising awareness about the benefits of ecological restoration and promoting sustainable practices.

Approaches to Ecological Restoration

1. **Passive Restoration:** Allowing natural processes to recover the ecosystem with minimal human intervention. This is often suitable for ecosystems that still have sufficient resilience and natural regenerative capacity.
2. **Active Restoration:** Directly intervening to restore the ecosystem, often necessary for severely degraded areas. This can include planting, soil treatment, and water management.
3. **Rewilding:** Reintroducing native species, including apex predators and keystone species, to restore natural processes and trophic interactions.

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4. **Agroecological Restoration:** Integrating restoration principles into agricultural landscapes to enhance biodiversity, soil health, and ecosystem services while maintaining agricultural productivity.
5. **Urban Ecological Restoration:** Restoring natural habitats within urban areas to enhance biodiversity, provide recreational spaces, and improve environmental quality for urban residents.

Ecological restoration is a multidisciplinary effort that requires collaboration among ecologists, land managers, policymakers, and the community. It plays a crucial role in combating biodiversity loss, mitigating climate change, and ensuring the sustainability of natural resources.

Social Forestry

Social forestry, also known as community forestry, is a forestry management approach that involves local communities in the planning, implementation, and management of forest resources. The primary goal is to improve the livelihoods of local people while ensuring the sustainable management of forest resources. This approach integrates social, economic, and environmental objectives to benefit both communities and ecosystems. Here are the key aspects and benefits of social forestry:

Key Aspects of Social Forestry

1. **Community Participation:**
 - **Involvement:** Local communities, including indigenous groups, farmers, and other stakeholders, are actively involved in decision-making processes.
 - **Empowerment:** Communities are empowered with knowledge, skills, and resources to manage forest resources effectively.
2. **Sustainable Management:**
 - **Conservation:** Protecting and conserving forest ecosystems to maintain biodiversity, water resources, and soil fertility.
 - **Utilization:** Sustainable harvesting of forest products such as timber, non-timber forest products (NTFPs), and medicinal plants.
3. **Economic Development:**
 - **Income Generation:** Creating opportunities for local communities to generate income through the sale of forest products, eco-tourism, and other forest-based activities.
 - **Employment:** Providing employment opportunities in forest management, conservation projects, and value-added processing of forest products.
4. **Social Benefits:**
 - **Poverty Alleviation:** Contributing to poverty reduction by providing alternative livelihoods and enhancing food security.
 - **Community Cohesion:** Strengthening social bonds and cooperation within communities through collective management efforts.
5. **Environmental Education:**
 - **Awareness:** Raising awareness about the importance of forests and sustainable practices through education and outreach programs.
 - **Training:** Providing training on sustainable forestry practices, agroforestry, and conservation techniques.

Benefits of Social Forestry

1. **Environmental Benefits:**
 - **Biodiversity Conservation:** Protecting and enhancing biodiversity by preserving natural habitats and restoring degraded ecosystems.
 - **Climate Mitigation:** Contributing to climate change mitigation through carbon sequestration and reducing deforestation.
 - **Soil and Water Conservation:** Improving soil health and water retention, which benefits agriculture and reduces erosion.
2. **Economic Benefits:**
 - **Diversified Income:** Providing a diversified source of income for rural communities, reducing dependence on single crops or external employment.
 - **Market Access:** Facilitating access to markets for forest products, which can improve local economies.
3. **Social Benefits:**
 - **Empowerment and Capacity Building:** Enhancing the capacity of local communities to manage resources and improve their livelihoods.
 - **Improved Livelihoods:** Enhancing food security and nutritional status by integrating agroforestry and other sustainable practices.

Examples of Social Forestry Programs

1. **Joint Forest Management (JFM) in India:** A collaborative approach where local communities and forest departments work together to manage forest resources, share benefits, and ensure sustainable use.
2. **Community Forests in Nepal:** Communities are granted legal rights to manage and benefit from forest resources, leading to improved forest conditions and livelihoods.
3. **Agroforestry Systems:** Integrating trees and shrubs into agricultural landscapes to enhance productivity, diversify income, and improve environmental sustainability.
4. **Participatory Forest Management in Africa:** Programs in countries like Tanzania and Ethiopia that involve local communities in forest conservation and sustainable use.

Challenges in Social Forestry

1. **Resource Conflicts:** Disputes over land and resource use can arise between communities and other stakeholders.
2. **Capacity Building:** Ensuring that communities have the necessary skills and knowledge to manage forests sustainably.
3. **Policy and Legal Frameworks:** Developing supportive policies and legal frameworks that recognize and protect community rights and interests.
4. **Sustainability:** Balancing immediate economic needs with long-term sustainability goals.

Social forestry represents a holistic approach to forestry management that recognizes the vital role of local communities in maintaining forest health and productivity. By integrating social, economic, and environmental objectives, social forestry can contribute significantly to sustainable development and environmental conservation.

Types of Ecosystem

An **ecosystem** (or **ecological system**) is a system that environments and their organisms form through their interaction. The biotic and abiotic components are linked together through nutrient cycles and energy flows.

- Ecosystems are controlled by external and internal factors. External factors such as climate, parent material which forms the soil and topography, control the overall structure of an ecosystem but are not themselves influenced by the ecosystem. Internal factors are controlled, for example, by decomposition, root competition, shading, disturbance, succession, and the types of species present.
- An ecosystem is a community of living and non-living things that interact with each other in a specific environment.
- Living things (biotic factors): Plants (Producers), animals (Consumers), micro-organisms (Decomposers).
- Non-living things (abiotic factors): water, air, soil, sunlight, temperature, rocks, minerals.
- Ecosystems can be small (a puddle of water) or large (a desert or oceans) and they can be terrestrial (land based) or aquatic (water based).
- Ecosystem is for managing natural resources, conserving biodiversity, and addressing environmental challenges like climate change and pollution.

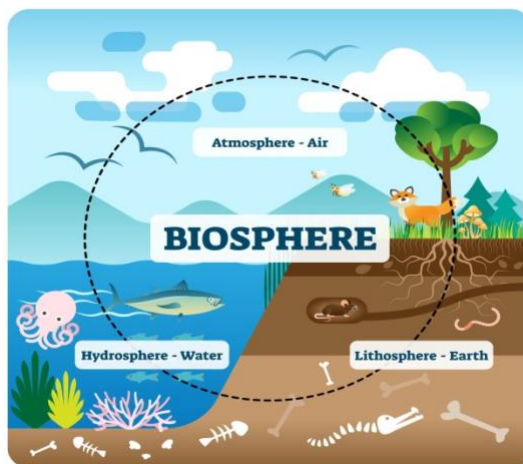


Fig: Ecosystem

The term "ecosystem" was first used in 1935 in a publication by British ecologist Arthur Tansley. The term was coined by Arthur Roy Clapham, who came up with the word at Tansley's request. Tansley devised the concept to draw attention to the importance of transfers of materials between organisms and their environment. He later refined the term, describing it as "The whole system, including not only the organism-complex, but also the whole complex of physical factors forming what we call the environment". Tansley regarded ecosystems not simply as natural units, but as "mental isolates". Tansley later defined the spatial extent of ecosystems using the term "ecotope".

Types of Ecosystem are as follows:



1. Forest Ecosystem:



Fig: The Amazon rainforest alongside the Solimoes River, a tropical rainforest

A **forest** is an ecosystem characterized by a dense community of trees. Hundreds of definitions of forest are used throughout the world, incorporating factors such as tree density, tree height, land use, legal standing, and ecological function. The United Nations' Food and Agriculture Organization (FAO) defines a forest as, "Land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 percent, or trees able to reach these thresholds *in situ*. It does not include land

that is predominantly under agricultural or urban use." Using this definition, *Global Forest Resources Assessment 2020* (FRA 2020) found that forests covered 4.06 billion hectares (10.0 billion acres; 40.6 million square kilometers; 15.7 million square miles), or approximately 31 percent of the world's land area in 2020.

Forests are the largest terrestrial ecosystems of Earth by area, and are found around the globe. 45 percent of forest land is in the tropical latitudes. The next largest share of forests are found in subarctic climates, followed by temperate, and subtropical zones.

Forests account for 75% of the gross primary production of the Earth's biosphere, and contain 80% of the Earth's plant biomass. Net primary production is estimated at 21.9 gigatonnes of biomass per year for tropical forests, 8.1 for temperate forests, and 2.6 for boreal forests.

Forests form distinctly different biomes at different latitudes and elevations, and with different precipitation and evapotranspiration rates. These biomes include boreal forests in subarctic climates, tropical moist forests and tropical dry forests around the Equator, and temperate forests at the middle latitudes. Forests form in areas of the Earth with high rainfall, while drier conditions produce a transition to savanna. However, in areas with intermediate rainfall levels, forest transitions to savanna rapidly when the percentage of land that is covered by trees drops below 40 to 45 percent. Research conducted in the Amazon rainforest shows that trees can alter rainfall rates across a region, releasing water from their leaves in anticipation of seasonal rains to trigger the wet season early. Because of this, seasonal rainfall in the Amazon begins two to three months earlier than the climate would otherwise allow. Deforestation in the Amazon and anthropogenic climate change hold the potential to interfere with this process, causing the forest to pass a threshold where it transitions into savanna.

Deforestation threatens many forest ecosystems. Deforestation occurs when humans remove trees from a forested area by cutting or burning, either to harvest timber or to make way for farming. Most deforestation today occurs in tropical forests. The vast majority of this deforestation is because of the production of four commodities: wood, beef, soy, and palm oil. Over the past 2,000 years, the area of land covered by forest in Europe has been reduced from 80% to 34%. Large areas of forest have also been cleared in China and in the eastern United States, in which only 0.1% of land was left undisturbed. Almost half of Earth's forest area (49 percent) is relatively intact, while 9 percent is found in fragments with little or no connectivity. Tropical rainforests and boreal coniferous forests are the least fragmented, whereas subtropical dry forests and temperate oceanic forests are among the most fragmented. Roughly 80 percent of the world's forest area is found in patches larger than 1 million hectares (2.5 million acres). The remaining 20 percent is located in more than 34 million patches around the world – the vast majority less than 1,000 hectares (2,500 acres) in size.



Fig: Deforestation of the Amazon rainforest in Brazil's Maranhão state, 2016

Human society and forests can affect one another positively or negatively. Forests provide ecosystem services to humans and serve as tourist attractions. Forests can also affect people's health. Human activities, including unsustainable use of forest resources, can negatively affect forest ecosystems.

Forest ecosystems broadly differ based on climate; latitudes 10° north and south of the equator are mostly covered in tropical rainforest, and the latitudes between 53°N and 67°N have boreal forest. As a general rule, forests dominated by angiosperms (*broadleaf forests*) are more species-rich than those dominated by gymnosperms (*conifer, montane, or needle leaf forests*), although exceptions exist. The trees that form the principal structural and defining component of a forest may be of a great variety of species (as in tropical rainforests and temperate deciduous forests), or relatively few species over large areas (e.g., taiga and arid montane coniferous forests). The biodiversity of forests also encompasses shrubs, herbaceous plants, mosses, ferns, lichens, fungi, and a variety of animals.

Types of Forests:

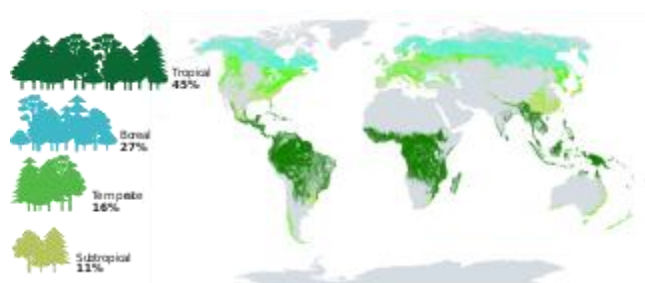


Fig: Proportion and distribution of global forest area by climatic domain, 2020

Forests are classified differently and to different degrees of specificity.

- I. The Tropical forests
- II. The Temperate forests
- III. The Boreal or Taiga forests

One such classification is in terms of the biomes in which they exist, combined with leaf longevity of the dominant species (whether they are evergreen or deciduous). Another

distinction is whether the forests are composed predominantly of broadleaf trees, coniferous (needle-leaved) trees, or mixed.

- Boreal forests occupy the subarctic zone and are generally evergreen and coniferous.
- Temperate zones support both broadleaf deciduous forests (e.g., temperate deciduous forest) and evergreen coniferous forests (e.g., temperate coniferous forests and temperate rainforests). Warm temperate zones support broadleaf evergreen forests, including laurel forests.
- Tropical and subtropical forests include tropical and subtropical moist forests, tropical and subtropical dry forests, and tropical and subtropical coniferous forests.
- Forests are classified according to physiognomy based on their overall physical structure or developmental stage (e.g. old growth vs. second growth).
- Forests can also be classified more specifically based on the climate and the dominant tree species present, resulting in numerous different forest types (e.g., Ponderosa pine/Douglas fir forest).

The number of trees in the world, according to a 2015 estimate, is 3 trillion, of which 1.4 trillion are in the tropics or sub-tropics, 0.6 trillion in the temperate zones, and 0.7 trillion in the coniferous boreal forests. The 2015 estimate is about eight times higher than previous estimates, and is based on tree densities measured on over 400,000 plots. It remains subject to a wide margin of error, not least because the samples are mainly from Europe and North America.



Fig: Broadleaf forest in Bhutan

2. Desert Ecosystem



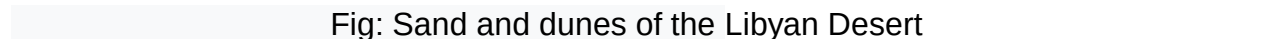


Fig: Sand and dunes of the Libyan Desert

A **desert** is a landscape where little precipitation occurs and, consequently, living conditions create unique biomes and ecosystems. The lack of vegetation exposes the unprotected surface of the ground to denudation. About one-third of the land surface of the Earth is arid or semi-arid. This includes much of the polar regions, where little precipitation occurs, and which are sometimes called polar deserts or "cold deserts". Deserts can be classified by the amount of precipitation that falls, by the temperature that prevails, by the causes of desertification or by their geographical location.^[1]

Deserts are formed by weathering processes as large variations in temperature between day and night put strains on the rocks, which consequently break in pieces. Although rain seldom occurs in deserts, there are occasional downpours that can result in flash floods. Rain falling on hot rocks can cause them to shatter, and the resulting fragments and rubble strewn over the desert floor are further eroded by the wind. This picks up particles of sand and dust, which can remain airborne for extended periods – sometimes causing the formation of sand storms or dust storms. Wind-blown sand grains striking any solid object in their path can abrade the surface. Rocks are smoothed down, and the wind sorts sand into uniform deposits. The grains end up as level sheets of sand or are piled high in billowing sand dunes. Other deserts are flat, stony plains where all the fine material has been blown away and the surface consists of a mosaic of smooth stones, often forming desert pavements, and little further erosion takes place. Other desert features include rock outcrops, exposed bedrock and clays once deposited by flowing water. Temporary lakes may form and salt pans may be left when waters evaporate. There may be underground sources of water, in the form of springs and seepages from aquifers. Where these are found, oases can occur.



Fig: Sahara Desert

Plants and animals living in the desert need special adaptations to survive in the harsh environment. Plants tend to be tough and wiry with small or no leaves, water-resistant cuticles, and often spines to deter herbivory. Some annual plants germinate, bloom and die in the course of a few weeks after rainfall, while other long-lived plants survive for years and have deep root systems able to tap underground moisture. Animals need to keep cool and find enough food and water to survive. Many are nocturnal, and stay in the shade or underground during the heat of the day. They tend to be efficient at conserving water, extracting most of their needs from their food and concentrating their urine. Some animals remain in a state of dormancy for long periods, ready to become active again during the rare rainfall. They then reproduce rapidly while conditions are favorable before returning to dormancy.

People have struggled to live in deserts and the surrounding semi-arid lands for millennia. Nomads have moved their flocks and herds to wherever grazing is available, and oases have provided opportunities for a more settled way of life. The cultivation of semi-arid regions encourages erosion of soil and is one of the causes of increased desertification. Desert farming is possible with the aid of irrigation, and the Imperial Valley in California provides an example of how previously barren land can be made productive by the import of water from an outside source. Many trade routes have been forged across deserts, especially across the Sahara, and traditionally were used by caravans of camels carrying salt, gold, ivory and other goods. Large numbers of slaves were also taken northwards across the Sahara. Some mineral extraction also takes place in deserts, and the uninterrupted sunlight gives potential for the capture of large quantities of solar energy.



Fig: The world's largest non-polar deserts

Deserts occupy about one third of Earth's land surface. Bottomlands may be salt-covered flats. Eolian processes are major factors in shaping desert landscapes. Polar deserts (also seen as "cold deserts") have similar features, except the main form of precipitation is snow rather than rain. Antarctica is the world's largest cold desert (composed of about 98% thick continental ice sheet and 2% barren rock). Some of the barren rock is to be found in the so-called Dry Valleys of Antarctica that almost never get snow, which can have ice-encrusted saline lakes that suggest evaporation far greater than the rare snowfall due to the strong katabatic winds that even evaporate ice.

Types of Deserts:

I. Subtropical Desert:



Fig: Subtropical Types of Desert

They are the hottest of all deserts found between 15-30 degrees both to the north and south of the equator along the Tropic of Cancer and Tropic of Capricorn, respectively. These deserts experience extreme temperature conditions and very low rainfall.

Temperature: Reaches above 50°C during the day, which decreases very sharply

during the night, owing to low humidity.

Yearly rainfall: Quite scarce, as the weather could be so extreme that raindrops evaporate before reaching the ground. Inland areas of Sahara receive less than 1.5 cm rainfall, which is one of the driest places on the planet.

Soil type: Coarse-textured and rocky with no subsurface water.

Plants that grow here: Mainly shrubs and small trees which are adapted to prevent loss of moisture; examples include creosote bush, mesquite tree, yuccas, ocotillo, sotol, ephedras, and agaves.

Animals found here: Nocturnal, and mainly carnivorous, animals here hide during hot daytime hours, becoming active during the night-time when it gets colder. Examples include mammals like camels, insects, reptiles, arachnids, bandicoot, kangaroo rat, and desert iguana.

Examples of subtropical deserts: Sahara desert in Africa is the largest of all subtropical deserts. Some other deserts of this group include Chihuahuan, Sonoran, and Mojave in the USA, the Kalahari Desert of Southern Africa, and the Tanami Desert of Northern Australia.

II. Cold Winter Desert



Fig: Cold Winter Types of Desert

They are also referred to as the semi-arid desert, located near the edges of other

deserts and in the continental dry areas. These deserts are characterized by long dry summer and mild, wet winters.

Temperature: Between 21-27°C in summer, which can go up to 38°C; evenings are cooler by about 10 degrees. During winter, the temperature ranges between 2-4°C, which can go down to 40°C.

Annual rainfall: Between 2-4 cm in average

Soil type: Sandy, fine-textured with loose rock fragments and low salt content

Plants that grow here: Mainly, cactus, as well as creosote bush, bur sage, whitethorn, mesquite, lyceums, and jujube. The leaves of cacti are transformed into spines to reduce loss of water.

Animals found here: Mammals like kangaroo and rabbits; insects like grasshoppers and ants; reptiles like lizards and snakes; birds like burrowing owls and California thrasher. During day time, these animals hide in the shadows of shrubs or in the burrows to protect themselves from intense heat.

Examples of cold winter deserts: Gobi desert in China and Mongolia, Patagonian desert in Argentina, Great Basin in the USA.

III. Coastal Desert



Fig: Coastal Types of Desert

Coastal deserts are located on the western coasts of the continents found between 20-30-degree latitude. These are characterized by cold winters and relatively long and

warm summers.

Temperature: Between 13-24°C during summer, with maximum range being about 35°C. During winter, it stays around 5°C or lower.

Annual rainfall: About 8-13 centimeters, which is more, compared to both the subtropical and coastal deserts.

Soil type: Fine-textured with a high content of salt and other nutrients.

Plants that grow here: Those growing here have roots that are close to the surface to absorb rainwater. Examples include saltbush, buckwheat bush, little leaf horsebrush, black sage, and *Chrysothamnus*.

Animals found here: Mammals like coyote and badger, amphibians like toads, birds like great horned owl, golden eagle, and bald eagle, reptiles like lizards and snakes, and Insects.

Examples of coastal deserts: The Atacama Desert in South America (driest place in the planet), the Namib Desert, and The Western Sahara desert of Africa.

IV. Polar Desert



Fig: Polar Types of Desert

Polar deserts are located in the Arctic and Antarctic regions. They are characterized by long cold winters and short warm summers. Winter witnesses relatively higher rainfall than summer.

Temperature: Ranges between 21-26°C (during summer), and between 2-4°C in winter.

Yearly rainfall: Between 15-26 cm in average

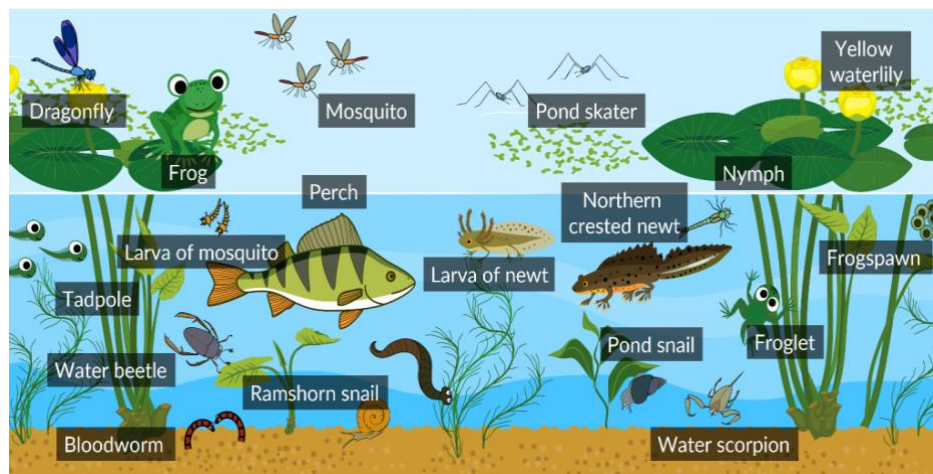
Soil type: Salty and contains silt. Soil is porous, where the alluvial fan is found.

Plants that grow here: Mostly algae, lichens, and some fungi. Short grasses, birches, willows, and mosses might grow in moist areas.

Animals found here: Jackrabbits, kangaroo rats, kangaroo mice, pocket mice, grasshopper mice, antelope, and ground squirrels.

Examples of Polar desserts: The Antarctic is the world's largest polar desert, which extends over New Zealand, Australia, France, Norway, the United Kingdom, Chile, and Argentina. The second-largest polar desert, the Arctic, spans across Alaska, Canada, Greenland, Iceland, Sweden, Russia, and Finland.

3. Marine Ecosystem:



Marine ecosystems are the biggest ecosystems. They cover around 71% of the total earth's surface. The marine ecosystem contains around 97% of the total water present on earth. Marine water contains a high amount of salts and minerals. Saltwater supports different types of species than freshwater. Marine ecosystems are the most abundant types of ecosystems in the world. Marine ecosystems include oceans, tidal zones, estuaries, salt marshes, saltwater swamps, mangroves, and coral reefs.

Marine ecosystems are the largest of Earth's aquatic ecosystems and exist in waters that have a high salt content. These systems contrast with freshwater ecosystems, which have a lower salt content. Marine waters cover more than 70% of the surface of

the Earth and account for more than 97% of Earth's water supply and 90% of habitable space on Earth. Seawater has an average salinity of 35 parts per thousand of water. Actual salinity varies among different marine ecosystems. Marine ecosystems can be divided into many zones depending upon water depth and shoreline features. The oceanic zone is the vast open part of the ocean where animals such as whales, sharks, and tuna live. The benthic zone consists of substrates below water where many invertebrates live. The intertidal zone is the area between high and low tides. Other near-shore (neritic) zones can include mudflats, seagrass meadows, mangroves, rocky intertidal systems, salt marshes, coral reefs, lagoons. In the deep water, hydrothermal vents may occur where chemosynthetic sulfur bacteria form the base of the food web.

Marine ecosystems are characterized by the biological community of organisms that they are associated with and their physical environment. Classes of organisms found in marine ecosystem include brown algae ,corals, cephalopods, echinoderms, and sharks.

Marine ecosystems are important sources of ecosystem services and food and jobs for significant portions of the global population. Human uses of marine ecosystems and pollution in marine ecosystems are significantly threats to the stability of these ecosystems. Environmental problems concerning marine ecosystems include unsustainable exploitation of marine resources (for example overfishing of certain species), marine pollution, climate change, and building on coastal areas. Moreover, much of the carbon dioxide causing global warming and heat captured by global warming are absorbed by the ocean, ocean chemistry is changing through processes like ocean acidification which in turn threatens marine ecosystems.



Fig: Coral reefs form complex marine ecosystems with tremendous biodiversity.

Because of the opportunities in marine ecosystems for humans and the threats created by humans, the international community has prioritized "Life below water" as Sustainable Development Goal 14. The goal is to "Conserve and sustainably use the oceans, seas and marine resources for sustainable development".¹

Types of Marine Ecosystem:

a) Coral reefs:



Fig: Coral reef

Coral reefs are one of the most well-known marine ecosystems in the world, with the largest being the Great Barrier Reef. These reefs are composed of large coral colonies of a variety of species living together. The corals form multiple symbiotic relationships with the organisms around them.

b) Mangroves:



Fig: Mangrove forests

Mangroves are trees or shrubs that grow in low-oxygen soil near coastlines in tropical or subtropical latitudes. They are an extremely productive and complex ecosystem that connects the land and sea. Mangroves consist of species that are not necessarily related to each other and are often grouped for the characteristics they share rather than genetic similarity. Because of their proximity to the coast, they have all developed adaptations such as salt excretion and root aeration to live in salty, oxygen-depleted water. Mangroves can often be recognized by their dense tangle of roots that act to protect the coast by reducing erosion from storm surges, currents, wave, and tides. The mangrove ecosystem is also an important source of food for many species as well as excellent at sequestering carbon dioxide from the atmosphere with global mangrove carbon storage is estimated at 34 million metric tons per year.

c) Seagrass meadows:



Fig: Seagrass meadow

Seagrasses form dense underwater meadows which are among the most productive ecosystems in the world. They provide habitats and food for a diversity of marine life comparable to coral reefs. This includes invertebrates like shrimp and crabs, cod and flatfish, marine mammals and birds. They provide refuges for endangered species such as seahorses, turtles, and dugongs. They function as nursery habitats for shrimps, scallops and many commercial fish species. Seagrass meadows provide coastal storm protection by the way their leaves absorb energy from waves as they hit the coast. They keep coastal waters healthy by absorbing bacteria and nutrients, and slow the speed of climate change by sequestering carbon dioxide into the sediment of the ocean floor.

Seagrasses evolved from marine algae which colonized land and became land plants, and then returned to the ocean about 100 million years ago. However, today seagrass meadows are being damaged by human activities such as pollution from land runoff, fishing boats that drag dredges or trawls across the meadows uprooting the grass, and overfishing which unbalances the ecosystem. Seagrass meadows are currently being destroyed at a rate of about two football fields every hour.

d) Kelp forests:



Fig: Kelp forest

Kelp forests occur worldwide throughout temperate and polar coastal oceans. In 2007, kelp forests were also discovered in tropical waters near Ecuador.

Physically formed by brown macroalgae, kelp forests provide a unique habitat for

marine organisms and are a source for understanding many ecological processes. Over the last century, they have been the focus of extensive research, particularly in trophic ecology, and continue to provoke important ideas that are relevant beyond this unique ecosystem. For example, kelp forests can influence coastal oceanographic patterns and provide many ecosystem services.

However, the influence of humans has often contributed to kelp forest degradation. Of particular concern are the effects of overfishing near shore ecosystems, which can release herbivores from their normal population regulation and result in the overgrazing of kelp and other algae. This can rapidly result in transitions to barren landscapes where relatively few species persist. Already due to the combined effects of overfishing and climate change, kelp forests have all but disappeared in many especially vulnerable places, such as Tasmania's east coast and the coast of Northern California. The implementation of marine protected areas is one management strategy useful for addressing such issues, since it may limit the impacts of fishing and buffer the ecosystem from additive effects of other environmental stressors.

e) Estuaries:



Fig: Estuaries

Estuaries occur where there is a noticeable change in salinity between saltwater and freshwater sources. This is typically found where rivers meet the ocean or sea. The wildlife found within estuaries is unique as the water in these areas is brackish - a mix of freshwater flowing to the ocean and salty seawater. Other types of estuaries also exist and have similar characteristics as traditional brackish estuaries. The Great Lakes are a prime example. There, river water mixes with lake water and creates freshwater estuaries. Estuaries are extremely productive ecosystems that many humans and animal species rely on for various activities. This can be seen as, of the 32 largest cities in the world, 22 are located on estuaries as they provide many environmental and economic benefits such as crucial habitat for many species, and being economic hubs for many coastal communities. Estuaries also provide essential ecosystem services such as water filtration, habitat protection, erosion control, gas regulation nutrient cycling, and it even gives education, recreation and tourism opportunities to people.

f) Salt marsh:



Fig: Salt marshes

Salt marshes are a transition from the ocean to the land, where fresh and saltwater mix. The soil in these marshes is often made up of mud and a layer of organic material called peat. Peat is characterized as waterlogged and root-filled decomposing plant matter that often causes low oxygen levels (hypoxia). These hypoxic conditions causes growth of the bacteria that also gives salt marshes the sulfurous smell they are often known for. Salt marshes exist around the world and are needed for healthy ecosystems and a healthy economy. They are extremely productive ecosystems and they provide essential services for more than 75 percent of fishery species and protect shorelines from erosion and flooding. Salt marshes can be generally divided into the high marsh, low marsh, and the upland border. The low marsh is closer to the ocean, with it being flooded at nearly every tide except low tide. The high marsh is located between the low marsh and the upland border and it usually only flooded when higher than usual tides are present. The upland border is the freshwater edge of the marsh and is usually located at elevations slightly higher than the high marsh. This region is usually only flooded under extreme weather conditions and experiences much less waterlogged conditions and salt stress than other areas of the marsh.

4. Grassland Ecosystem:



Fig: Nachusa Grasslands, spring 2016

A **grassland** is an area where the vegetation is dominated by grasses (Poaceae). However, sedge (Cyperaceae) and rush (Juncaceae) can also be found along with variable proportions of legumes, like clover, and other herbs. Grasslands occur naturally on all continents except Antarctica and are found in most ecoregions of the Earth. Furthermore, grasslands are one of the largest biomes on Earth and dominate the landscape worldwide.^[1] There are different types of grasslands: natural grasslands, semi-natural grasslands, and agricultural grasslands.^[1] They cover 31–69% of the Earth's land area.

Grasslands dominated by unsown wild-plant communities ("unimproved grasslands") can be called either natural or "semi-natural" habitat. Although their plant communities are natural, their maintenance depends upon anthropogenic activities such as grazing and cutting regimes. The semi-natural grasslands contain many species of wild plants, including grasses, sedges, rushes, and herbs; 25 plant-species per 100 square centimeters can be found. A European record that was found on a meadow in Estonia described 76 species of plants in one square meter. Chalk downlands in England can support over 40 species per square meter.

Classifications of grassland:

Grassland types by Schimper (1898, 1903):

- Meadow (hygrophilous or trophophilous grassland)
- Steppe (xerophilous grassland)
- Savannah (xerophilous grassland containing isolated trees)



Fig: Steppe family: a common grassland animal, the swift fox

Grassland types by Ellenberg and Mueller-Dombois (1967):

Formation-class V. Terrestrial herbaceous communities

1. Savannas and related grasslands (tropical or subtropical grasslands and parklands)
2. Steppes and related grasslands (e.g. North American "prairies" etc.)
3. Meadows, pastures or related grasslands
4. Sedge swamps and flushes
5. Herbaceous and half-woody salt swamps



Fig: Forb vegetation A hike through the Tallgrass Prairie Heritage Park in Canada

Grassland types by Laycock (1979):

1. Tallgrass (true) prairie
2. Shortgrass prairie
3. Mixed-grass prairie
4. Shrub steppe
5. Annual grassland
6. Desert (arid) grassland
7. High mountain grassland

General Types of Grasslands Ecosystem:

i. Tropical and subtropical:

These grasslands can be classified as the tropical and subtropical grasslands, savannas and shrublands biome. The rainfall level for that grassland type is between 90 and 150 centimeters per year. Grasses and scattered trees are common for that ecoregion, as well as large mammals, such as wildebeest (*Connochaetes taurinus*) and zebra (*Equus zebra*). Notable tropical and subtropical grasslands include the Llanos grasslands of South America.



Fig: Cumberland Plain Woodland, a grassy woodland that covers Western Sydney

ii. Temperate:

Mid-latitude grasslands, including the prairie and Pacific grasslands of North America, the Pampas of Argentina, Brazil and Uruguay, calcareous downland, and the steppes of Europe. They are classified with temperate savannas and shrublands as the temperate grasslands, savannas, and shrublands biome. Temperate grasslands are the home to many large herbivores, such as bison, gazelles, zebras, rhinoceroses, and wild horses. Carnivores like lions, wolves, cheetahs and leopards are also found in temperate grasslands. Other animals of this region include deer, prairie dogs, mice, jack rabbits, skunks, coyotes, snakes, foxes, owls, badgers, blackbirds, grasshoppers, meadow larks, sparrows, quails, hawks and hyenas.

iii. Flooded:

Grasslands that are flooded seasonally or year-round, like the Everglades of Florida, the Pantanal of Brazil, Bolivia and Paraguay or the Esteros del Ibera in Argentina, are classified with flooded savannas as the flooded grasslands and savannas biome and occur mostly in the tropics and subtropics. The species that live in these grasslands are well adapted to the hydrologic regimes and soil conditions. The Everglades—the world's largest rain-fed flooded grassland—is rich in 11,000 species of seed-bearing plants, 25 species of orchids, 300 bird species, and 150 fish species.

Water-meadows are grasslands that are deliberately flooded for short periods.



Fig: Grassland in the Antelope Valley, California

iv. Montane:

High-altitude grasslands located on high mountain ranges around the world, like the Páramo of the Andes Mountains. They are part of the montane grasslands and shrublands biome and can be tropical, subtropical, and temperate. The plants and animals, that can be found in the tropical montane, are able to adapt to cool, wet conditions as well as intense sunlight.

v. Desert and xeric:

Also called desert grasslands, they are composed of sparse grassland ecoregions located in the deserts and xeric shrublands biome. Temperature extremes and low amount of rainfall characterize these kinds of grasslands. Therefore, plants and animals are well adapted to minimize water loss.

BIOTIC AND ABIOTIC COMPONENTS OF ECOSYSTEM

INTRODUCTION

Ecosystem: It is basic functional unit of Ecology.

A group of organisms interacting among themselves and with environment is known as ecosystem. Thus an ecosystem is a community of different species interacting with one another and with their non living environment exchanging energy and matter.

Ecology is the study of ecosystems.

Example:

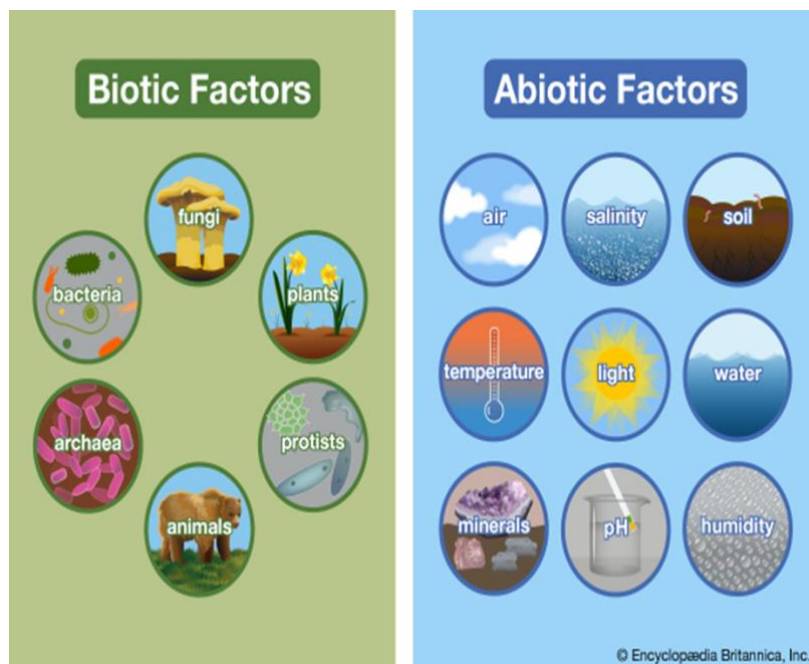
Animals cannot synthesis their food directly but depend on the plants either directly or indirectly.

STRUCTURE OF AN ECOSYSTEM

The term structure refers to the various components of an ecosystem

An ecosystem has two major components

- Biotic (Living) Components
- Abiotic (Non-Living) Components



BIOTIC COMPONENTS

It includes all living organisms and their products. This group includes all animals, plants, bacteria, fungi and their waste products like fallen leaves or branches or excreta. The biotic components of ecosystem can be classified as flora and fauna based on their structure and other features.

Functionally the living organisms can be classified as,

- Producers
- Consumers
- Decomposers

PRODUCERS

Producers are called as autotrophs. They can be two main types,

- Photosynthetic forms
- Chemosynthetic forms

The photosynthetic forms are green plants that convert solar energy into chemical energy, this is called Photosynthesis. The producers therefore include green plants, photosynthesis bacteria and chemosynthetic bacteria. On land photosynthesis is carried out mainly by higher plants.

In the sea the main photosynthetic organisms are the microscopic algae, planktons, diatoms and the flagellates.



CONSUMERS

Heterotrophic organisms are unable to synthesize their own food and hence obtain them by feeding on other organisms. The decomposers also fit in to this definition.

The consumers are classed into various categories based on the nature of the food they consume. Such as Herbivores, Carnivores and Omnivores

And also be classified as,

- Primary consumers
- Secondary consumers
- Tertiary consumers

1.PRIMARY CONSUMERS

Primary consumers are herbivores, which feed on plant material. The amount they consumed are commonly referred to as the consumption rate. Based on these the ecosystem can be grouped as high rated, low rated.

Primary consumers can be grazers or browsers.



Plant eating insect

2.SECONDARY CONSUMERS

Heterotrophic animals which feed on herbivorous organisms or primary consumers are termed as the secondary consumers.

These animals therefore carnivorous.



3.TERTIARY CONSUMERS

These are carnivorous heterotrophs that feed on other carnivorous animals.Top carnivorous are few in number.

Most birds of pray and cats fit this category.



Hawk

DECOMPOSERS

Decomposers feed on dead material,and that is first broken down before being absorbed.The detritivores ,plays the initial role of breaking up large bodies in to small particles.

Decomposers are mainly fungi and bacteria.



Fungus



Bacteria

ABIOTIC COMPONENTS

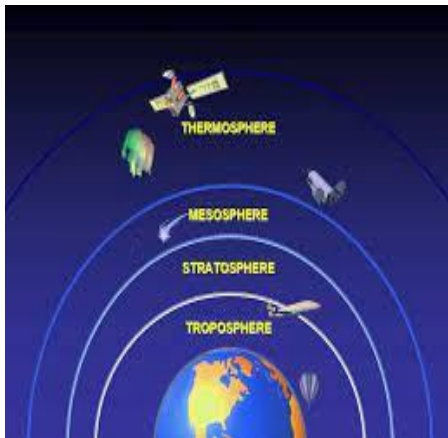
Abiotic components or abiotic factors are non-living chemical and physical parts of the environment that affect living organisms and the functioning of ecosystems.It include physical conditions and non-living resources that affect living organisms in terms of growth, maintenance, and reproduction

The abiotic portion of the biosphere consists of three parts .They are the

- Atmosphere (Air)
- Hydrosphere (The water)
- Lithosphere (The solid earth)

ATMOSPHERE

- The Earth's atmosphere is a thin envelope of gases surrounding the solid planet, the hydrosphere, and biosphere. This can be differentiated on the basis of temperature into four layers.
- They are The Troposphere, The stratosphere, The mesosphere and The thermosphere.
- In relation to the ecosystem the most important layer is the troposphere; which is the lowest stratum. It is roughly 10km in thickness and is usually thicker in equatorial regions than at the poles.
- The ozone layer in the stratosphere, absorbs ultra violet radiation, thus preventing it from reaching the surface of the earth.



THE ROLE OF TROPOSPHERE

- All organisms obtain their requirement of gasses from the troposphere which is composed of 78% Nitrogen, 21% Oxygen, and 0.03% Carbon dioxide, and less of others such as Hydrogen, Helium and Methane.
- It determines the weather phenomena.
- Formation of the clouds.
- Changes in troposphere results in changes of climatic factors such wind and ocean currents.
- These currents of air and water in turn strongly influence the distribution of precipitation, both in time and in space.
- Water in the troposphere is warmed both by the solar energy and the heat radiating from the earth's surface. part of this heat is radiated back to earth. this leads to the retention of heat, keeping the earth relatively warm even at night when there is temporarily no flux of solar energy.

THE HYDROSPHERE

- Water is one of the most unusual natural compounds found on earth.
- It is essential to all life as both an internal and an external medium
- Water is one of the main agents in the weathering of rocks which is necessary for the formation of soil, and erosion, deposition of sediment, the process that shape land forms.
- Water covers 71% of the earth's surface and is the medium for several different ecosystems.
- The water molecule has polar covalent bonds and thus causes ionic compounds to dissociate and dissolve in it.-The universal solvent.



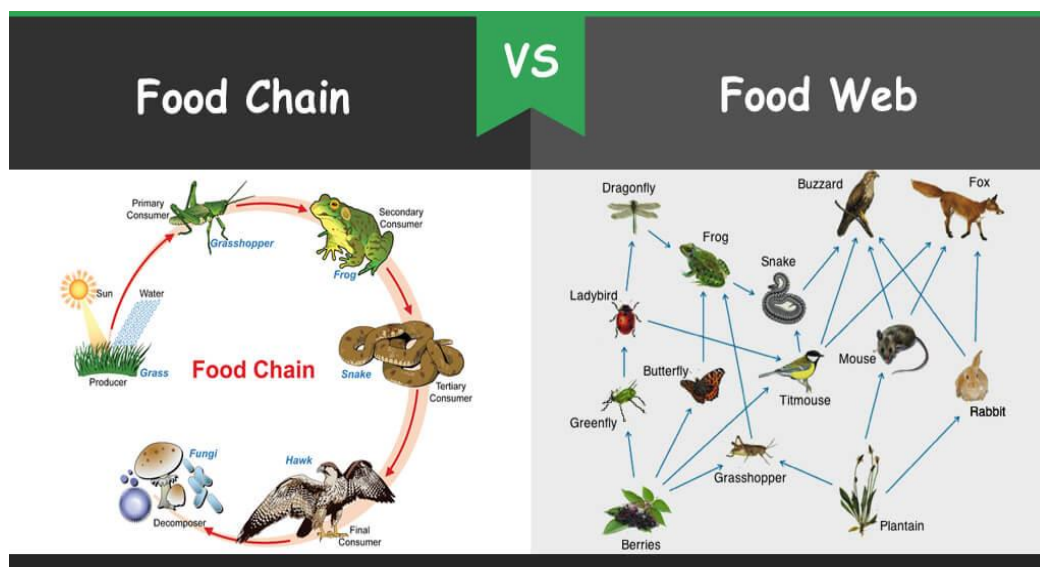
THE LITHOSPHERE

- The lithosphere can be divided into three main components. Rocks, Sediments and Soil .
- Rocks can be defined as consolidated units of the earth's crust which are formed of minerals that have come together by hardening.
- The role of rocks include facilitating storage and movement of ground water. serve as sources of mineral constituents of sediment and soils.
- Weathering
- The distribution and nature of rocks determine the types and distribution of ecosystems.
- Sediments can be defined as rock fragments that may not be chemically altered by weathering.
- Soils consists of complex mixture of rock fragments, highly altered minerals, organic debris and living organisms.

- The importance of soil to the ecosystem lies in that they are the source of almost all nutrients and much of the water available to the organisms in terrestrial ecosystems.
- Soils are the medium for the detritus food chain, and support all organisms.



FOOD CHAIN AND FOOD WEB



FOOD CHAIN

There sequence of eating and being eaten in an ecosystem is known as food chain or transfer of food energy from the plants through a series of organisms is known as food chain.

- A food chain always starts with plant life and ends with animal. When the organisms die, they are all decomposed by micro-organism (bacteria and fungi) into nutrients that can again be used by the plants.
- At each and every level, nearly 80-90% of the potential energy gets lost as heat.

Trophic Levels (T1, T2, T3, T4, T5). The various steps through which food energy passes in an ecosystem is called as trophic levels. The trophic levels are arranged in the following way,

- The green plants or producers represent first trophic level T1.
- The herbivores or primary consumers represent second trophic level T2.
- The carnivores or secondary consumers represent third trophic level T3.
- The tertiary consumers are fourth trophic level T4.
- Finally decomposers represent last trophic level T5.

Food chains are classified into two main types

- Grazing food chain
- Detritus food chain

GRAZING FOOD CHAIN : Found in grassland ecosystem and pond ecosystems. Grazing food chain starts with green plants (primary producers) and goes to decomposer food chain or detritus food chain through herbivores and carnivores.

DETRITUS FOOD CHAIN: Found in grassland ecosystems and forest ecosystems. Detritus food chain starts with dead organic matter (plants and animals) and goes to decomposer food chain through herbivores and carnivores.

FOOD WEB

The interlocking pattern of various food chain in an ecosystem is known as food web.

- In a food web many food chains are interconnected, where different types of organisms are connected at different trophic levels, so that there are number of opportunities of eating and being eaten at each trophic level.
- Grass may be eaten by insects, rats, deer's etc.. these may be eaten by carnivores (snake, fox, tiger). Thus there is an interlocking of various food chains called food webs.

DIFFERENCE BETWEEN FOOD CHAIN AND FOOD WEB

In a linear food chain if one species gets affected or becomes extinct, then the species in the subsequent trophic levels are also affected. But, in a food web if one species gets affected, it doesn't affect other trophic levels so seriously. There are number of options available at each trophic level.

SIGNIFICANCE OF FOOD CHAIN AND FOOD WEB

- Food chains and food webs play a very important role in the ecosystem. Energy flow and nutrient cycling takes place through them.
- They maintain and regulate the population size of different trophic levels, and thus help in maintaining ecological balance.
- They have the property of bio-magnification. The non-biodegradable materials keep on passing from one trophic level to another. At each successive trophic level, the concentration keeps on increasing. This process is known as bio-magnification.